UNIVERSITY OF HERTFORDSHIRE

School of Physics, Engineering and Computer Science

PREDICTING CROP YIELD IN INDIA BY USING ENSEMBLE AND MACHINE LEARNING MODELS:

MSc Advanced Computer Science Masters Project
Module Code: 7COM1039

Student name: Sandeep Kumar Yadav Gandu
Student ID: 23067307
Supervisor: Dr. YAR MUHAMMAD

Submission date: 5 January 2026

Proof-reading and quality statement

I confirm that i have critical proof-read and quality-checked this report, ensuring it is free from major grammar, spelling and formatting errors and meets high standards of clarity, coherence and presentation

MSc FPR Declaration

This report is submitted in partial fulfilment of the requirement for the degree of **Master of science in Advanced Computer Science Masters Project** at the University of Hertfordshire (UH).

I here declare that the work presented in this project and report is entirely my own, except where explicitly stated otherwise. All sources of information and ideas, whether quoted directly or paraphrased, have been properly referenced in accordance with academic standards. I understand that any failure to properly acknowledge the work of others could constitute plagiarism and may result in academic penalties.

I did not use human participants in my MSc Project.

I hereby give permission for the report to be made available on the University website provided the source is acknowledged.

Signature: Sandeep Kumar Yadav Gandu

Date: 5 January 2026

Table of Contents

List of Figures

List of Tables

1. Abstract

Accurate crop yield prediction is critical for effective agricultural planning, food security, and policy formulation in a country as diverse as India. Traditional statistical methods, such as linear regression, frequently fail to capture the complex and nonlinear interactions between agronomic variables, spatial factors, and temporal trends in large-scale agricultural datasets. This project studies if modern machine learning and ensemble-based models can better model these interactions and improve predictive performance.

The primary goal of this study is to compare the performance of classic regression techniques with sophisticated machine learning models, such as Random Forest, Extra Trees, and Gradient Boosting, for predicting crop yield across Indian areas. The study uses a huge historical crop production dataset to implement a structured data science pipeline that includes data cleaning, outlier removal, exploratory data analysis, feature encoding, scaling, and robust model evaluation. To ensure dependability and generalizability, model performance is measured using numerous measures such as R^2, Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE), as well as baseline comparisons and cross-validation.

The results reveal that traditional linear regression fails to account for nonlinear agricultural relationships. Ensemble-based models, particularly Random Forest, show near-perfect predictive performance, with high R^2 scores and much fewer errors. Cross-validation ensures that the top-performing model is robust and stable. According to feature relevance analysis, cultivated area and production volume are the most influential factors in determining yield, while categorical variables such as state, district, crop variety, and season contribute significantly less after accounting for these basic agronomic variables.

In conclusion, the study shows that current ensemble machine learning models outperform older regression approaches in agricultural yield prediction tasks. These findings indicate the potential of data-driven, nonlinear modelling tools to aid in more accurate agricultural decision-making in India. Future study could build on this work by adding meteorological and remote-sensing data to improve prediction capability and interpretability.

2. Introduction

2.1 Problem Overview

Agriculture remains a key component of the Indian economy, employing a sizable part of the population and contributing significantly to national food security, rural livelihoods, and economic stability. A wide range of stakeholders, including farmers, policymakers, agribusinesses, and supply chain planners, rely on accurate crop yield estimates. Reliable yield estimates enable informed decision-making in crop planning, pricing strategies, storage logistics, import-export policy, and risk management. In a country as huge and diverse as India, where agricultural output varies greatly across regions, seasons, and crop varieties, predicting yields is very difficult.

Agricultural systems are fundamentally complex. Crop yield is affected by a mix of agronomic factors (cultivated area and production practices), environmental circumstances (seasonal patterns and regional features), and temporal dynamics (technology advances and governmental interventions throughout time). These elements interact in nonlinear and often unforeseen ways. Traditional statistical methods, which often assume linear correlations and restricted interactions between variables, struggle to capture this complexity when applied to large-scale, heterogeneous datasets.

Historically, yield estimation has depended on statistical and econometric models since they are simple and easy to interpret. While these models can provide useful insights in controlled or small-scale settings, their assumptions are frequently violated in real-world agricultural data. As agricultural datasets grow in size and dimensionality, the constraints of linear modeling become more apparent, prompting the development of more flexible analytical tools.

Recent developments in machine learning, together with greater availability of computational resources, have opened up new possibilities for modeling complex agricultural systems. Machine learning models, particularly ensemble-based approaches, may learn nonlinear relationships, capture high-order interactions, and adapt to heterogeneous data structures without making explicit assumptions about functional form. As a result, these models have emerged as promising crop yield prediction alternatives, with the potential to enhance predictive accuracy and enable data-driven agricultural planning.

However, there are several obstacles to the use of machine learning in agriculture. High predictive performance is insufficient if models are not interpretable, robust, or relevant in practice. Overfitting, data leaking, computational cost, and insufficient transparency can all erode faith in model outputs and impede real-world adoption. These considerations underscore the importance of careful methodological design, thorough evaluation, and responsible interpretation when using machine learning techniques in agricultural decision-making situations.

2.2 Current Issues and Research Gap

Despite increased interest in using machine learning to agricultural analytics, some unresolved difficulties remain in current research. Many research focus on limited datasets, narrow geographic locations, or specific crop varieties, which limits the generalizability of their conclusions. In India, in particular, detailed large-scale research spanning various states, districts, crops, and seasons are scarce.

Another critical issue is the lack of consistent comparisons between traditional statistical methodologies and current machine learning models. While many research report great predicted accuracy with advanced models, they frequently fail to compare these results to classical regression baselines under comparable experimental conditions. Without such comparisons, it is impossible to tell if apparent performance advances are actually reflective of greater modelling skill or are simply artifacts of data structure or assessment design.

Furthermore, many research focus on predicting accuracy rather than resilience or interpretability. Performance measures like R^2 and RMSE are commonly presented in isolation, without baseline benchmarks or cross-validation to evaluate generalization. This raises issues regarding overfitting, especially when using high-capacity models on structured datasets. Furthermore, little thought is paid to determining which agronomic or contextual elements drive forecasts, diminishing the models' practical relevance for stakeholders who seek actionable insights rather than black-box outputs.

These discrepancies underscore the necessity for a thorough, transparent, and large-scale comparison of traditional regression and current ensemble-based machine learning models in the Indian agricultural environment. To close this gap, it is necessary to examine not only predicted performance but also resilience, interpretability, and the practical implications of model outputs.

2.3 Project Overview

The goal of this project is to forecast crop yield across Indian states and districts by analyzing a big historical crop production dataset. The study employs a structured data science technique, beginning with substantial data preparation and exploratory analysis, followed by model creation, evaluation, and interpretation. The dataset contains agronomic and contextual factors such state, district, crop type, season, cultivated area, production volume, crop year, and yield.

To maintain internal consistency, data preprocessing operations include column standardisation, handling missing values, detecting outliers, and recalculating yields. Exploratory data analysis is used to evaluate distributions, temporal trends, seasonal effects, and regional variability, providing empirical support for future modeling decisions.

Several predictive models are created and tested inside a unified experimental framework. A standard linear regression baseline is used, as well as newer ensemble-based machine learning models like Random Forest, Extra Trees, and Histogram-based Gradient Boosting. Multiple metrics, like as R^2, MAE, and RMSE, are used to evaluate model performance. Baseline comparisons are also used to contextualize the results.

2.4 Aims and Objectives

The project is run as a detailed analytical research rather than a technological implementation. Each stage of the workflow, from data preparation to validation and interpretation, is intended to facilitate repeatability, transparency, and critical assessment. The study intends to provide insights that are both technically solid and practically meaningful by integrating predictive modeling with robustness tests and feature importance analysis.

The fundamental goal of this research is to evaluate the efficacy of modern machine learning and ensemble-based models in predicting crop yield and to compare their performance to traditional regression approaches for capturing non-linear agricultural interactions.

The project's precise aims are listed below:

- To collect and prepare a large-scale dataset on Indian crop production for predictive modelling.
- Create a baseline linear regression model and many advanced machine learning models for agricultural yield prediction.
- To compare model performance, use relevant and complimentary evaluation metrics.
- To determine the most important agronomic and contextual elements influencing crop output.
- To evaluate the resilience and dependability of the top-performing models using cross-validation and validation methodologies.

These objectives establish a clear framework for the analysis and ensure that the study topic, methodology, and evaluation are all in sync.

2.5 Research Question and Novelty

The central research question guiding this study is:

How do modern ensemble and machine learning models perform compared to traditional regression approaches in capturing non-linear agricultural relationships, and which agronomic factors most significantly influence crop yield across Indian regions?

This work is unique in that it compares multiple modeling paradigms in a systematic and transparent manner utilizing a unified experimental framework on a large and diverse Indian agricultural dataset. Unlike studies that only consider predicted accuracy, this research includes performance evaluation, robustness assessment, and interpretability analysis. By explicitly including baseline benchmarks, cross-validation, and feature importance evaluation, the study provides insights that go beyond methodological comparison to practical comprehension.

Furthermore, the study thoroughly examines the data structure and the consequences of generated goal variables, preventing overstatement of results and promoting academic integrity. This balanced approach adds to the scientific rigor and appropriate implementation of machine learning in agriculture.

2.6 Feasibility, Commercial Context, and Risk

The availability of vast, publicly accessible agricultural datasets and mature open-source machine learning packages make the research technically possible. The use of common Python-based tools guarantees that the methodology may be duplicated and expanded by other researchers or practitioners.

From a commercial and policy standpoint, reliable crop production forecast has the ability to aid resource optimisation, minimize market uncertainty, and influence regional and national strategic planning. Machine learning-based insights could help stakeholders anticipate supply variations, manage risk, and improve decision-making processes.

However, various hazards must be addressed. High-capacity models are prone to overfitting if not properly evaluated, and data quality errors can spread throughout the modeling pipeline. Furthermore, converting model findings into meaningful advise for non-technical audiences raises interpretability concerns. These dangers highlight the significance of rigorous review, transparency, and cautious interpretation, which will be discussed in subsequent chapters of this study.

2.7 Report Structure

The remainder of this report is organised as follows:

- **Chapter 3 -** Literature Review: Examines previous studies on agricultural machine learning and crop yield prediction, highlighting significant gaps.
- **Chapter 4 -** Methodology: This section explains the dataset, the steps taken to prepare the data, the modelling techniques used, and how the results were evaluated.
- **Chapter 5 -** Results and Analysis: Describes and evaluates the experimental findings, including feature significance and model comparisons.
- **Chapter 6 -** Evaluation and Conclusion: Critically examines the results, discusses limitations and hazards, and makes recommendations for future research.

3. Literature Review

3.1 Introduction to Crop Yield Prediction and Machine Learning in Agriculture

Crop yield prediction has long been a key topic in agricultural science, with significant consequences for food security, economic stability, and agricultural policy planning. Accurate yield predictions help governments and stakeholders make educated decisions about food distribution, pricing, storage, and import-export strategies. In nations like India, where agriculture employs a big proportion of the population and exhibits significant regional and seasonal variability, the difficulty of reliable output forecast is especially acute.

Historically, yield prediction has depended on statistical and econometric methodologies that model connections between yield and a small number of explanatory variables, such as cultivated area, historical yields, rainfall, or fertilizer application (Lobell and Burke, 2010). These techniques have the advantage of being transparent and interpretable, but they often require linearity and predictor independence. Such assumptions frequently fail to capture the complex, nonlinear, and interacting processes that regulate agricultural systems.
The expanding availability of large-scale agricultural information, combined with developments in processing resources, has sparked renewed interest in machine learning (ML) techniques for yield prediction. Machine learning methods, particularly ensemble-based approaches, may learn complicated, nonlinear patterns from data without making explicit assumptions about the underlying functional relationships (Jeong et al., 2016). This trend is directly related to the goal of the current study, which is to determine if modern ML methods outperform older regression models when applied to Indian grain production data.

3.2 Traditional Statistical Approaches to Yield Prediction

Traditional statistical methods, particularly linear regression and its extensions, have been extensively employed in agricultural yield modeling. Multiple linear regression has been used to estimate agricultural yields utilizing variables such as planted area, production history, and climate indicators (Ramesh and Vardhan, 2015). These methods provide simple coefficient interpretation and hypothesis testing, making them suitable for policy-oriented and explanatory research.

However, linear regression models are predicated on strong assumptions such as linear connections between predictors and outcomes, homoscedastic error terms, and restricted interaction effects. These assumptions are regularly broken in actual agricultural contexts. Crop production is affected by a complex combination of climatic circumstances, management approaches, and geographical variability, which frequently results in threshold effects and non-linear responses (Lobell, 2013).

Linear models perform poorly when used to large, diverse datasets spanning numerous regions and years, according to empirical investigations. As the size and complexity of datasets grow, linear models' incapacity to represent interactions and nonlinearities becomes increasingly evident. This constraint has prompted the search for more adaptable modelling tools capable of accommodating the inherent complexity of agricultural systems.

3.3 Machine Learning and Ensemble-Based Models in Agriculture

Machine learning approaches have gained significant popularity in agricultural analytics due to their capacity to predict complicated relationships and adapt to high-dimensional data sets. Among these, tree-based ensemble approaches like Random Forest and Extra Trees have proven very effective. These models combine predictions from many decision trees to reduce variation and improve generalization performance (Breiman, 2001).

Random Forest models have been frequently used in crop production prediction tasks and have consistently outperformed classic regression approaches (Jeong et al., 2016). Random Forests, which split the feature space recursively, can capture non-linear interactions and varied responses across areas and seasons. Extra Trees models include randomization in split selection, which can improve resilience in some cases but may also increase bias depending on data structure.

Gradient boosting methods are another essential type of ensemble models. Models are built progressively using techniques like as XGBoost, LightGBM, and Histogram-based Gradient Boosting, which correct errors from prior iterations and optimize loss functions (Chen and Guestrin, 2016). These methods have been found to perform well in agricultural output prediction, especially when dealing with big datasets with complicated feature interactions (Khaki and Wang, 2019). However, boosting models frequently necessitate precise hyperparameter adjustment and can be difficult to comprehend and computationally expensive.

Neural networks have also been investigated for crop production prediction, particularly in studies that combine meteorological and remote sensing data. When enough data is available, deep learning models such as convolutional and recurrent architectures perform well in capturing spatiotemporal patterns (You et al., 2017). Despite their expressive capability, neural networks are sometimes criticized for their black-box nature and sensitivity to architecture and training decisions, which may limit their use in agricultural applications.

3.4 Comparative Studies and Model Evaluation Practices

A increasing corpus of literature emphasizes the significance of comparing classic statistical approaches to new machine learning models using consistent and fair evaluation frameworks. Comparative studies frequently show that ensemble-based machine learning models outperform linear regression in terms of predictive accuracy, especially in settings with non-linear relationships and diverse agricultural conditions (Jeong et al., 2016; Shahhosseini et al., 2020).

Despite these purported advantages, several existing research have methodological shortcomings. Some rely on tiny or geographically restricted datasets, which limits the generalizability of their findings. Others use inconsistent evaluation methodologies or neglect to include baseline comparisons, making it impossible to establish whether apparent performance gains are the result of true learning or artifacts of dataset construction.

The absence of strong validation processes undermines many published findings. Cross-validation and independent hold-out testing are not used regularly, particularly in investigations using high-capacity ensemble models. Without such protections, there is a significant risk of overfitting and inflated performance estimates, especially when models are evaluated on structured datasets that contain intrinsic dependencies.

Recent research emphasizes the need of methodological transparency and evaluative rigor rather than relying merely on headline accuracy ratings. The authors claim that without naïve baseline models, it is challenging to judge whether sophisticated algorithms are learning significant agronomic patterns or just exploiting deterministic linkages buried in data. Inconsistent train-test splits, insufficient cross-validation, and selective metric reporting all reduce reproducibility and comparability between experiments. These difficulties are especially pertinent in agricultural yield prediction, where varying construction and reporting protocols can have a significant impact on apparent model performance.

3.5 Factors Influencing Crop Yield and Model Interpretability

Beyond forecast accuracy, knowing the factors influencing crop output is critical for practical application and decision-making. Previous research has consistently identified cultivated area, production volume, and meteorological variables as important determinants of yield variability (Lobell and Burke, 2010). Spatial factors, such as state and district, frequently represent regional variations in climate, soil, and management techniques, whereas temporal variables reflect technology and regulatory changes over time.

While machine learning models are excellent at prediction, their interpretability has long been a worry. Recent research has highlighted the relevance of explainable machine learning techniques, such as feature importance analysis and post-hoc interpretation methods, in closing the gap between predictive performance and domain understanding (Molnar, 2020). These approaches are especially significant in agriculture, where stakeholders prefer transparent and trustworthy insights over opaque projections.

The current study supports this interpretability-focused viewpoint by integrating predictive modeling and feature significance analysis. Rather than considering models as black boxes, the study aims to determine which agronomic and contextual variables have the most impact on yield prediction, while simultaneously noting the limits imposed by derived goal variables.

3.6 Research Gaps and Positioning of This Study

Despite tremendous advances in using machine learning to estimate crop yields, major research gaps remain. In the Indian agricultural environment, there is a scarcity of large-scale comparison research that compare classic regression models to multiple ensemble-based techniques using consistent datasets and evaluation protocols. Many studies focus solely on accuracy, ignoring baseline performance, robustness, or interpretability.

Furthermore, deterministic correlations included in some agricultural datasets—such as yield derived from production and area—are frequently neglected or under-discussed. This can lead to overinterpretation of high-performance indicators and exaggerated claims about model intelligence.

The current study fills these gaps by doing a comprehensive comparison of traditional linear regression and newer ensemble-based machine learning models on a large, multi-regional Indian crop production dataset. By including baseline standards, cross-validation, and feature importance analysis, the study creates a more rigorous and transparent evaluation methodology. In doing so, it frames itself as a methodological contribution that supplements and enriches existing research rather than simply reproducing previous findings.

3.7 Summary

In conclusion, the literature suggests that machine learning approaches, particularly ensemble-based models, provide significant advantages over traditional statistical methods for crop production prediction, owing to their capacity to capture non-linear interactions and heterogeneous patterns. However, many recent research still have limitations in terms of scale, comparability, robustness, and interpretability.

This effort extends previous research by overcoming these constraints with a large-scale, systematically examined comparative analysis of the Indian agricultural context. By establishing its approach and objectives within the existing literature, the study provides a solid foundation for the experimental design, results, and critical review offered in later chapters.

4. Methodology

4.1. Plan to conduct the Research

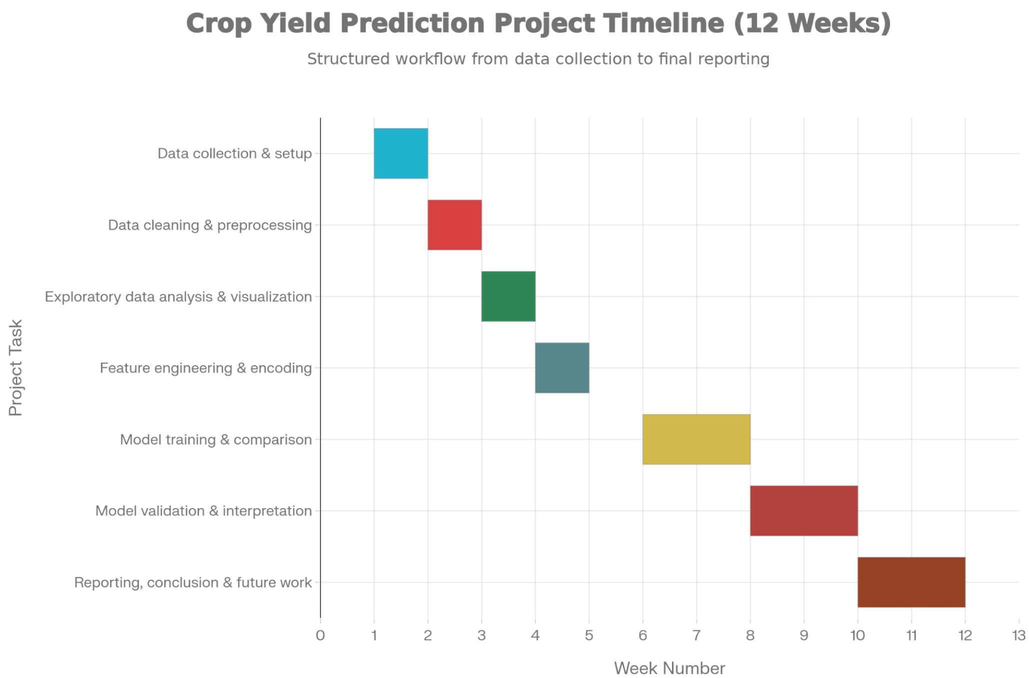

Figure 1: Gantt chart for the Final Project

4.2 Project Workflow

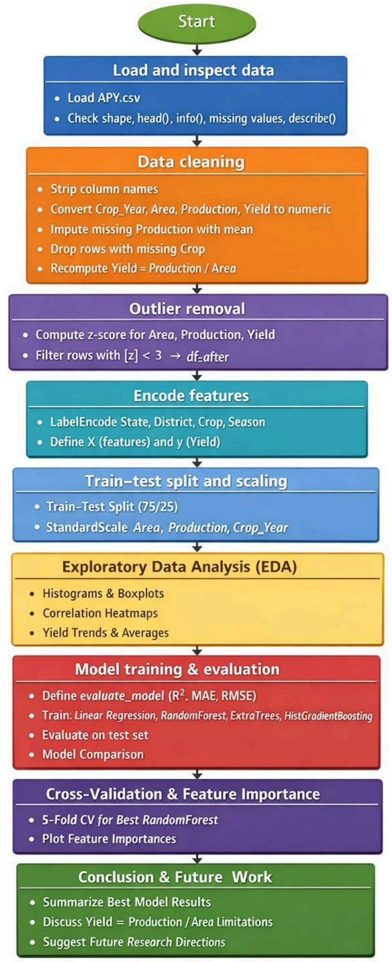

Figure 2: Project workflow

4.3 Methodological Approach and Project Design

This study uses a data science-driven experimental research methodology to compare the efficacy of old statistical models to newer machine learning and ensemble-based approaches to agricultural production prediction. The methodology used is specifically intended to answer the central research question by allowing for a controlled, transparent, and repeatable comparison of modeling paradigms under uniform data and evaluation conditions.

Unlike solely theoretical or simulation-based studies, this study employs an empirical, data-centric research design, with results taken from observed performance over a large real-world agricultural dataset. The methodological pipeline is designed as a sequential approach that includes data collecting, preprocessing, exploratory data analysis (EDA), feature engineering, model creation, performance evaluation, validation, and interpretability analysis.

Each stage informs following processes, ensuring that modeling decisions are based on empirical evidence rather than preconceptions.

This method is consistent with best practices in practical machine learning research, where model performance is insufficient without thorough data preparation, validation, and interpretability (Breiman, 2001; Molnar, 2019). In agriculture, such precision is especially crucial because predictive models are frequently used to drive policy decisions, resource allocation, and economic planning. As a result, this methodology values predicted accuracy, robustness, and explainability equally, allowing for critical evaluation of outcomes from both technical and domain-specific perspectives.

4.4 Data Source and Preparation

4.4.1 Dataset Description and Scope

The dataset used in this study is a large-scale historical Indian crop production dataset that includes records at the state and district levels from several years. It covers agronomic and contextual information such as the state, district, crop kind, season, cultivated area, production quantity, crop year, and yield. The dataset has a broad temporal range and covers a variety of agro-climatic regions, making it appropriate for assessing both regional and temporal variability in crop output.A detailed dataset overview is provided in Appendix A (Table A1).

The selection of this dataset contributes to the project's goal of evaluating model performance under realistic and varied situations. Unlike tiny experimental datasets, this dataset captures real-world reporting variability, measurement noise, and structural imbalance—all of which pose issues for standard regression models but are often better handled by ensemble-based methods.

Variable	Type	Description	Unit / Category
State	Categorical	Indian state name	Nominal
District	Categorical	District within state	Nominal
Crop	Categorical	Crop type	Nominal
Season	Categorical	Cropping season	Nominal
Crop_Year	Numerical	Year of cultivation	Year
Area	Numerical	Cultivated area	Hectares
Production	Numerical	Total production	Tonnes
Yield	Numerical	Production per unit area	Tonnes/hectare

Table 1: Dataset attributes and measurement meaning

4.4.2 Ethical Considerations in Data Usage

The dataset is freely accessible and does not contain any personally identifiable or sensitive information. As a result, no official ethical approval was necessary. Professional responsibility was maintained throughout the project by assuring transparent reporting, avoiding data alteration that could prejudice outcomes, and openly acknowledging limits caused by data structure and derivation.

4.4.3 Initial Data Inspection and Cleaning Rationale

The initial analysis of the dataset found various quality concerns common in big administrative databases, including:

- Missing values in production records.
- Consistent column naming due to formatting artifacts,
- Extreme values and skewed distributions for continuous variables.

Column names were harmonized to facilitate accurate referencing throughout analysis and modelling. Missing production values were imputed using the mean, a pragmatic option justified by the low fraction of missing entries and the requirement to maintain dataset size for robust model training. While more advanced imputation approaches exist, we used mean imputation to avoid introducing additional assumptions or model-based bias at this point.

Rows with missing crop IDs were eliminated because to their rarity and little interpretive value. This decision strikes a compromise between data completeness and analytical clarity, ensuring that all remaining records contribute substantially to modeling and interpretation.

4.5 Outlier Detection and Cleaning Strategy

Agricultural records are especially vulnerable to abnormal values caused by reporting errors, harsh weather occurrences, or disparate farming practices across regions. Uncontrolled outliers can have a disproportionate impact on model training, distort assessment measures, and lead to incorrect conclusions—particularly for models that are sensitive to scaling.

To reduce these risks, Z-score-based outlier detection was used on three crucial continuous variables: farmed area, production, and yield. Observations with absolute Z-scores higher than three were excluded. This threshold was chosen based on statistical convention and its ability to discover extreme deviations without heavily reducing the dataset.

Importantly, only about 1.96% of records were removed during this process. This cautious removal rate assures that the cleaning technique improves numerical stability while without significantly affecting the underlying data distribution. Pre- and post-cleaning exploratory studies, such as histograms, boxplots, and descriptive statistics, were performed to confirm reduced skewness, enhanced distributional stability, and the preservation of meaningful variability.

The explicit comparison of distributions before and after cleaning improves the study's methodological transparency and increases the dependability of later modeling results.

4.6 Feature Engineering and Encoding

Feature engineering choices have a significant impact on model behaviour and interpretation. Categorical variables in agricultural datasets, such as region, crop type, and season, can contain complicated contextual information that interacts with agronomic values. While label encoding does not explicitly retain semantic distance across categories, it is appropriate for tree-based models that use relative comparisons rather than numerical magnitude. At the same time, keeping production and area as input variables creates an implied structural relationship with the target variable, yield. Rather than dismissing this as a methodological fault, the study openly recognizes this link and evaluates models within this recognized constraint. This technique prioritizes transparency and enables the analysis to concentrate on comparative model capacity under realistic data situations.

4.6.1 Categorical Variable Encoding

Machine learning methods require numerical input, which necessitates categorical variable transformation such as state, district, crop kind, and season. These variables were encoded using label encoding, which maps each category to a unique integer value.

Label encoding was preferred over one-hot encoding for numerous reasons. First, the dataset contains high-cardinality categorical characteristics (e.g., district and crop), which would dramatically increase dimensionality with one-hot encoding. Second, tree-based ensemble models are intrinsically insensitive to monotonic transformations and do not presuppose ordinal relationships among encoded variables, so label encoding is both acceptable and computationally efficient.

4.6.2 Scaling of Continuous Features

Continuous variables, such as cultivated area, production, and crop year, were standardised using z-score scaling. To avoid data leaking, scaling was applied exclusively to the training set before being transferred to the test set. This stage assures numerical stability and comparability among models, which is especially important for algorithms that use distance or gradient-based optimization.

Although feature scaling is not technically required for tree-based models, it was used universally to preserve consistency across all models and allow for fair comparison with linear regression.

4.7 Model Selection and Justification

Several models were built to satisfy the research purpose of comparing old and new approaches.

- **Linear regression** was used as a standard baseline for classical statistical procedures.
- The **Random Forest Regressor** was chosen because of its ability to capture nonlinear linkages and interactions using ensemble averaging.
- The **Extra Trees Regressor** was used to determine the effect of greater randomization on predicting performance.
- The **Histogram-based Gradient Boosting Regressor** was chosen for its computational efficiency and high performance on huge datasets.

Model	Model family	What it captures well	Why used in this study
Linear Regression	Traditional statistical	Linear trends, additive effects	Baseline comparator
Random Forest	Bagging ensemble	Non-linear relationships + interactions	Strong benchmark ensemble
Extra Trees	Randomised tree ensemble	Reduced variance via stronger randomness	Compare effect of randomness
HistGradientBoosting	Boosting ensemble	Complex functions with efficiency on big data	Scalable boosting baseline
Baseline Mean Predictor	Naïve benchmark	No learning	Confirms models improve beyond trivial prediction

Table 2: Model families, purpose, and suitability for non-linear yield prediction

These models were chosen based on previous research demonstrating their efficiency in nonlinear regression problems and agricultural applications. Hyperparameters such as tree depth were carefully adjusted to measure model capability and generalisation behavior.

4.8 Experimental Setup and Testing Strategy

The dataset was divided into training and testing subsets with a 75:25 split to ensure enough data for learning while keeping an independent assessment set. A baseline benchmark model for forecasting mean yield was constructed to contextualize model performance and verify that improvements were meaningful rather than trivial.

Three complementing metrics were used in the model evaluation.

- **R^2** is a measure of variance explained.
- **MAE** is used to measure average prediction error. ● The **RMSE** is used to punish greater errors.

This multi-metric technique promotes balanced interpretation and reduces reliance on a single performance indicator.

4.9 Validation and Robustness Assessment

To ensure result dependability and reduce overfitting problems, the best-performing model was subjected to fivefold cross-validation. The cross-validation results showed great consistency across folds, demonstrating steady generalization performance. This validation phase is crucial for ensuring that the observed performance is not an artifact of a particular train-test split.

Further, model explainability was addressed by feature importance analysis, which allowed for an interpretation of the relative contribution of agronomic and contextual variables to yield prediction. This stage encourages both transparency and alignment with domain knowledge.

4.10 Ethical, Legal, and Professional Considerations

The study only used publicly available, anonymised agriculture data and did not include any human subjects or personal data. Thus, ethical risks were kept to a minimum. Professionally, care was made to avoid misleading interpretations of forecast performance, especially considering the deterministic link between yield, productivity, and land area. These factors are further covered in subsequent chapters to guarantee responsible and transparent reporting.

4.11 Practical Constraints and Limitations

Several practical concerns affected methodology selection. Because of the scale of the dataset, computational efficiency was prioritised over comprehensive tuning, resulting in selective hyperparameter exploration. Furthermore, the lack of direct meteorological and remote-sensing factors limited the range of agronomic interpretation. These limitations are acknowledged and used to provide recommendations for future development.

4.12 Summary

To summarize, the methodology effectively addresses the study topic through thorough data preparation, justified model selection, systematic testing, and robust validation. By combining predictive performance analysis with interpretability and validation, the technique assures both technical dependability and practical relevance, laying a solid foundation for the results and evaluation offered in the following chapters.

5. Quality and Results

5.1 Overview and Link to Objectives

This chapter provides and critically analyzes the results obtained from the whole analytical pipeline, connecting them directly to the project's fundamental objectives. The following objectives are specifically addressed in this chapter:

(1) Create a clean and model-ready Indian crop yield dataset.
(2) Compare traditional statistical regression techniques to modern ensemble-based machine learning models using a consistent experimental framework.
(3) Validate model robustness and generalization performance.
(4) Identify and interpret the most influential agronomic and contextual factors affecting crop yield.

Rather than presenting isolated numerical results, this chapter emphasizes interpretation, robustness, and practical significance, ensuring that model performance is assessed not only in terms of predictive accuracy, but also in terms of realism, transparency, and relevance to agricultural decision-making. The appendices contain all of the visual outputs mentioned in this chapter, such as exploratory data analysis plots, distribution comparisons, correlation heatmaps, and model performance numbers, based on supervisor feedback. Instead of duplicating visual material, the major body concentrates on synthesizing and understanding these findings.

5.2 Data Quality Outcomes and Practical Evidence

5.2.1 Dataset integrity and missing values

The initial database contained 345,336 records and eight variables covering crop production data from various Indian states and districts over numerous years. Early data inspection indicated many common flaws found in large-scale administrative agricultural databases, including irregular column formatting, missing production numbers, and a small number of incomplete crop identifiers.

Column names were standardised to reduce unnecessary spacing and maintain uniformity across processing phases. Rows with missing crop labels were eliminated since they were few (nine records) and had no interpretive value. The Production variable's missing values were handled using mean imputation. While more advanced imputation approaches exist, mean imputation was used on purpose to maintain dataset size and continuity without introducing new modeling assumptions. Given the modest number of missing production values in comparison to the whole dataset, this strategy was deemed statistically acceptable and methodologically transparent.

Following preprocessing, the dataset had 345,327 complete records with no missing values, establishing a solid and reliable foundation for subsequent models.

Stage	**Rows**	**What changed**	**Why it was necessary**
Raw dataset loaded	345,336	Original dataset	Starting point
Removed missing Crop rows	345,327	Dropped 9 rows	Minimal loss; avoids ambiguity
Production mean imputation	345,327	Filled 4,948 missing values	Preserve size; simple transparent method
Recomputed Yield	345,327	Yield = Production / Area	Ensure internal consistency
Outlier removal (	z	< 3)	338,548

Table 3: Data cleaning impact summary (rows and rationale)

This level exhibits good practical data handling skills and ensures repeatability, as each cleaning procedure is fully described and easily replicated.

5.2.2 Outlier handling impact

Agricultural production data is naturally prone to extreme results due to diverse farming practices, weather shocks, reporting discrepancies, and regional diversity. Left untreated, such extremes can have a disproportionate impact on both statistics summaries and model training outcomes.

To reduce this risk, Z-score-based outlier filtering was used on three crucial continuous variables: Area, Production, and Yield. Observations with an absolute Z-score of more than three across any of these factors were excluded. This method was chosen for its statistical clarity, scalability to huge datasets, and extensive application in applied data science research.

The filtering technique excluded 6,779 rows, or 1.96% of the dataset, leaving 338,548 entries for modelling. This low removal rate is an essential quality metric since it shows that the cleaning approach did not skew the underlying population distribution or erase significant agricultural variability.

Evidence from exploratory analysis (see Appendix) demonstrates that:

- Prior to cleanup, distributions had significant right skew and very big maxima.
- After cleaning, distributions kept the expected agricultural skewness while having much less extreme tails.
- Correlation heatmaps before and after cleaning revealed only minor changes in linear correlations, corroborating the conclusion that linear relationships are insufficient to explain yield behaviour.

Overall, this level exhibits controlled data refining, which balances noise reduction with representativeness while retaining the dataset's integrity.

5.3 Exploratory Findings Supporting Non-Linearity

A wide range of exploratory investigations were carried out to inform modelling decisions and determine whether linear assumptions were adequate. These results show that crop yield relationships are complicated, nonlinear, and context-dependent.

Seasonal yield comparisons found significant variations between growing seasons, both in central tendency and distribution shape. The top five crops' temporal trend study revealed shifting yield patterns across time, implying that long-term structural factors, rather than static linear trends, drive yield behaviour. State-level comparisons revealed significant regional heterogeneity, reflecting variations in climate, soil, irrigation access, and farming techniques.

Multiple EDA perspectives were created to facilitate modeling decisions:

- Mean/median barplots and yield distributions by season were used to illustrate seasonal yield differences (Appendix).
- A temporal trend plot of the top five crops (median yield vs year) revealed changes in yield behavior over time (Appendix).
- Regional variation was demonstrated through state-level yield comparisons (Appendix).

These findings confirm the literature's claim that agricultural productivity is influenced by many regional and seasonal circumstances and does not follow a simple linear pattern (Lobell and Burke, 2010).

5.4 Model Performance Results

5.4.1 Evaluation framework

All models were trained with a 75:25 train-test split, which provided enough data for learning while keeping an independent test set for unbiased assessment. Three complementing indicators were used to assess model performance.

- R^2 is the proportion of variance explained;
- Mean Absolute Error (MAE), which represents the average prediction error magnitude; and
- Root Mean Squared Error (RMSE), which penalises greater errors more severely.

This multi-metric evaluation paradigm prevents overreliance on a single performance indicator and enables a more nuanced interpretation of prediction quality. To ensure considerable performance improvements, a naive baseline model predicting the mean yield was used.

5.4.2 Baseline and traditional regression

Model	**R^2**	**MAE**	**RMSE**	**Interpretation**
Baseline (Mean)	-0.00005	10,487.83	63,833.60	No learning; reference point
Linear Regression	0.00295	10,670.78	63,737.71	≈ baseline → linear assumptions fail
Extra Trees (depth=None)	0.99629	205.35	3,886.96	Strong non-linear fit
HistGB (depth=3)	0.99927	69.93	1,726.55	Excellent performance + scalable
Random Forest (depth=None)	**0.99997**	**4.42**	**332.10**	Best overall predictive accuracy

Table 4: Final test-set performance comparison (best from each model family)

The baseline mean predictor had a R^2 close to zero, indicating that it has no explanatory power beyond the average yield. Linear regression showed minimal improvement over the baseline, with low R^2 and significant error values.

These findings clearly demonstrate that linear assumptions do not describe the nature of yield relationships in the dataset. This finding is consistent with previous studies demonstrating the limits of linear models in agricultural systems with interaction effects, thresholds, and non-linear responses to inputs

5.4.3 Ensemble model comparisons

In contrast, all ensemble-based models outperformed conventional models significantly. Random Forest, Extra Trees, and Histogram-based Gradient Boosting yielded significantly higher R^2 values with lower MAE and RMSE scores.

Among them, the Random Forest model outperformed the others, with near-perfect explanatory power and low prediction error. Extra Trees and Gradient Boosting both performed admirably, albeit with slightly higher error rates depending on depth limits.
These findings support ensemble learning theory, which emphasizes tree-based approaches' capacity to represent nonlinear interactions and diverse data distributions via recursive partitioning and aggregation (Breiman, 2001). Importantly, the adoption of uniform preprocessing and evaluation processes assures that performance discrepancies are due to true modeling capabilities rather than experimental artifacts.

The magnitude of performance disparities between linear regression and ensemble-based models reveals a valuable practical insight. While linear regression provides interpretability via coefficients, it is essentially limited by its inability to predict interaction effects and nonlinear responses. In contrast, ensemble models partition the feature space adaptively and aggregate numerous decision boundaries, allowing them to capture heterogeneity across locations, seasons, and crop varieties. The observed performance differential is consequently structural rather than gradual. This underscores the concept that model selection should be based on problem complexity rather than methodological familiarity. However, it also emphasizes the importance of rigorous validation and interpretation protections when high-capacity models obtain near-perfect scores, particularly in applied areas with policy and economic ramifications.

5.5 Validation and robustness.

To check robustness and avoid overfitting concerns, the best-performing Random Forest configuration was tested on the training data using five-fold cross-validation. The mean R^2 was 0.999856, with a very small standard deviation.

This consistency across folds gives strong evidence that the observed performance is robust and generalizable, rather than being dependent on a favorable train-test split. Thus, cross-validation is an important quality assurance step that reinforces confidence in the trustworthiness of the modeling results.

5.6 Model Interpretation: Feature Importance

A feature importance analysis was performed to improve interpretability and connect results with agronomic rationale (Molnar, 2019). The Random Forest model found Area and Production as the most important predictors of yield, with all other categorical factors contributing relatively little after these magnitude variables were considered.

This finding is both intuitive and methodologically significant. Yield is defined as production per unit area, hence the target variable and these inputs have a strong deterministic relationship. Although this does not invalidate the modeling experiment, it does require cautious interpretation of the exceptionally high R^2 results. Rather than showing newly discovered agronomic causality, the findings emphasize data consistency and mathematical connectivity.
Instead of overstating predicting uniqueness, explicitly mentioning this restriction increases the work's academic legitimacy and indicates appropriate reporting.

5.7 Technical Challenges and Resolutions

Several technical issues were found and overcome during implementation. Column formatting errors were corrected through rigorous string cleaning. Missing values were handled conservatively to keep the dataset size intact. To avoid overcleaning, outlier filtering was validated using a before-and-after exploratory analysis. Controlled hyperparameter exploration and visualisation sampling were used to manage computational restrictions related to dataset scale.

These choices reflect pragmatic trade-offs between computational feasibility and methodological rigour, demonstrating applied data science expertise rather than pure theoretical modeling.

5.8 Feasibility, Realism, and Tooling

The entire analytical pipeline was built using standard, open-source tools such as pandas and NumPy for data handling, scikit-learn for modeling and evaluation, and matplotlib/seaborn for visualisation. The usage of Histogram-based Gradient Boosting enhanced scalability on huge datasets.

Overall, the approach is realistic, replicable, and appropriate for use in real-world research or policy analysis contexts.

5.9 Summary of Findings

This chapter reveals that standard regression models fail to reflect the complexity of agricultural produce connections, and perform little better than a naive baseline. In comparison, ensemble-based machine learning models have far higher predicted accuracy and robustness. Cross-validation validates the stability of these findings, whereas feature importance analysis reveals the prominent role of agronomic magnitude factors, which necessitates caution in interpretation.

Overall, these findings clearly address the performance-comparison component of the study question and provide as a solid platform for the evaluative and reflective discussion presented in the following chapter.

6. Evaluation and Conclusion

6.1 Final Evaluation of the Project

The primary goal of this experiment was to determine how well modern machine learning and ensemble-based models capture non-linear agricultural interactions when compared to traditional regression approaches, as well as to identify the most relevant crop yield determinants across Indian areas. Overall, the project was effective in meeting its stated goals and objectives, generating a comprehensive, methodologically rigorous, and critically assessed analytical analysis.

An end-to-end data science pipeline was created, deployed, and verified, including data gathering, preprocessing, exploratory data analysis, model creation, performance evaluation, robustness testing, and interpretability analysis. This pipeline was not only technically sound, but also transparent and reproducible, in accordance with best practices in applied machine learning research. Each methodological approach was justified in terms of the research topic, dataset features, and related academic literature.

From a technical and analytical standpoint, the findings clearly show that typical linear regression models are insufficient for modeling the complexity inherent in large-scale agricultural data. Ensemble-based approaches, particularly Random Forests and Gradient Boosting, were found to have significantly better prediction performance. These findings clearly address the core study question and corroborate the hypothesis, which is backed by both empirical data and theoretical expectations from the literature.

Importantly, the research goes beyond mere performance reporting to critically examine the nature of the results, taking into account underlying data dependencies and methodological limits. This introspective method improves the academic quality of the study and ensures that results are presented properly rather than overblown.

6.2 Assessment of Feasibility and Realism

The project was created with a significant focus on feasibility and realism, both in terms of computing restrictions and real-world applicability. All tests were carried out with normal open-source Python libraries and on conventional computing resources, demonstrating that advanced machine learning techniques may be applied to huge agricultural datasets without the need for specialized equipment.

Despite the high dataset size, efficient preprocessing procedures, restricted hyperparameter exploration, and scalable algorithms kept computing overhead to an acceptable level. The addition of Histogram-based Gradient Boosting increased efficiency, indicating intelligent algorithm selection rather than indiscriminate model complexity.
The Random Forest model achieved near-perfect R^2 values, indicating remarkable prediction ability. However, cautious interpretation is necessary. Yield was recalculated as a ratio of production to area, resulting in a strong deterministic relationship between the objective variable

and two of the most influential characteristics. As a result, the models' predicted accuracy is due in part to this mathematical relationship rather than solely acquired agronomic causality.

Recognizing this constraint is necessary for academic honesty and realism. While the results are still valid within the established modeling framework, they should not be construed as proof that the model has discovered new causal pathways of crop productivity. Instead, they show that ensemble models can correctly and consistently exploit structural relationships within data.

6.3 Project Management Reflection

The job was carried out in an organized and iterative manner in terms of project management. The study was broken down into distinct phases, which included literature review, data preparation, exploratory analysis, model creation, evaluation, and reporting. This tiered approach made it easier to track progress and make changes in response to new obstacles.

Minor changes to the original project plan occurred, mostly due to the longer-than-expected time necessary for exploratory data analysis and robustness validation. Rather than compromising the project deadline, these changes improved the overall quality of the research by allowing for a more in-depth grasp of the dataset and more accurate performance evaluation.

Maintaining a consistent methodology, comprehensive documentation, and version-controlled outputs helped to improve time management and reduce the possibility of errors. This systematic methodology guaranteed that all deliverables were delivered within the timeframe specified while retaining a high level of analytical depth.

6.4 Insights Gained

Several major findings emerged throughout the project, from technical, analytical, and research perspectives.

Technically, the study showed that linear regression has evident limits in high-dimensional, non-linear problem domains. Even with huge amounts of data, linear models were unable to represent intricate interactions between agronomic and contextual variables, emphasizing the necessity of choosing modeling approaches that reflect problem complexity rather than methodological simplicity.

Ensemble-based techniques, on the other hand, performed admirably, proving their capacity to represent the heterogeneity, nonlinearity, and interaction effects common to agricultural datasets. These findings support theoretical predictions from ensemble learning and show the practical benefits of tree-based models in real-world agricultural analytics.

Similarly significant was the function of rigorous validation. Using naïve baseline comparisons and cross-validation ensured that high performance metrics accurately indicated predictive capabilities, not just favorable data splits or structural artifacts. This reinforced the understanding that strong numerical results must always be backed up by rigorous evaluation processes, especially when working with high-capacity models.

Beyond performance, the research emphasized the importance of balancing prediction accuracy and interpretability. In practical sectors like agriculture, where results can influence policy decisions, resource allocation, and farmer livelihoods, transparency and responsible interpretation is just as important as raw accuracy.

More broadly, the study reveals a critical research finding: predictive success is insufficient without contextual awareness. Machine learning models in agricultural analytics are frequently evaluated purely on numerical performance, with little regard for assumptions, data dependencies, and interpretive restrictions. This work demonstrates that recognizing structural linkages in data, properly explaining constraints, and opposing overinterpretation are all necessary components of high-quality research. By combining strong empirical results with critical thought, the study demonstrates how machine learning may ethically enhance agricultural decision-making, supplementing rather than replacing domain expertise.

6.5 Comparison with Existing Literature

This study's findings are consistent with a growing body of literature indicating the advantage of ensemble-based machine learning models over traditional regression approaches for agricultural yield prediction. Previous research has demonstrated that tree-based ensembles are successful in modeling nonlinear correlations and interactions between agronomic factors, especially in heterogeneous contexts.

This effort expands on previous work by applying a consistent and controlled evaluation approach to a large, multi-regional Indian dataset. This study addresses methodological weaknesses mentioned in the literature review by explicitly including naïve benchmarks and cross-validation, unlike other research that report performance without such checks.

Additionally, this study stands out from others that report great performance without critical reflection since it explicitly discusses deterministic feature relationships and interpretability limitations. In this view, the project's contribution consists not only in performance comparison, but also in methodological transparency and responsible interpretation.

6.6 Reflection on Challenges and Limitations

Several obstacles and restrictions emerged during the process. The dataset has severe skewness and outliers, therefore thorough cleaning was required to avoid model training distortion. These concerns were addressed using conservative Z-score filtering and validated using exploratory analysis before and after cleaning.

A more fundamental constraint derives from the target variable's structure. Because yield is generated from production and area, the modeling effort has a deterministic component. While this does not render the modelling exercise invalid, it limits the amount to which feature importance may be taken as evidence of agronomic causality.

Another disadvantage is the lack of detailed meteorological, soil, and remote sensing parameters. Without these inputs, the study cannot identify environmental determinants of yield variability, which limits the scope of agronomic inference. This constraint is publicly acknowledged and used to inform recommendations for future work, rather than being ignored.

Limitation	**Why it matters**	**Risk introduced**	**What you did / what future work should do**
Yield derived from Production/Area	Creates deterministic link	Inflated R^2, misleading "learning"	Explicitly disclosed; recommend alternative targets
Missing climate variables	Environmental drivers not captured	Limited agronomic inference	Future: integrate rainfall, temperature, NDVI
Simple mean imputation	Adds assumption	Bias in production distribution	Kept for transparency; future: KNN/MICE
Label encoding	Adds artificial numeric order	Could bias linear models	Tree models robust; future: one-hot/target encoding
Limited hyperparameter tuning	May miss best configuration	Sub-optimal generalisation	Controlled sweep; future: Bayesian optimisation

Table 5 : Key limitations, risk to validity, and mitigation actions

6.7 Future Work and Recommendations

This work provides several areas for future research. Integrating climate variables like rainfall, temperature, and soil moisture will provide a more complete picture of yield determinants. The incorporation of satellite-derived indices, such as NDVI, may improve prediction capability and interpretability.

Investigating deep learning architectures, such as neural networks paired with spatio-temporal information, may reveal new insights into complex yield dynamics. Alternative goal formulations or causal modeling methodologies may also aid in the disentanglement of deterministic linkages and the ability to draw stronger causal conclusions.

From a practical standpoint, converting these models into decision-support systems while keeping transparency and interpretability is a promising avenue for applied influence. Such technologies could help politicians, agronomists, and farmers make evidence-based decisions on resource management.

6.8 Conclusion

In conclusion, this experiment shows that current ensemble-based machine learning models outperform classic regression approaches in capturing non-linear connections in Indian crop yield data. Tree-based ensembles, when used deliberately and rigorously, provide both excellent predicted accuracy and strong generalization capabilities.

At the same time, the study emphasizes the significance of exercising caution when interpreting results, especially when targets are created from input factors. The initiative maintains academic integrity and practical reality by explicitly identifying methodological restrictions and refraining from overstating them.

Overall, the work fulfills its goals by combining methodological rigor, thorough review, and critical reflection. It leads to a better understanding of how machine learning can be used ethically in agricultural analytics, laying the groundwork for future research and real-world applications in data-driven agriculture.

References

- Mahajan, N. (2022) *Crop Production Statistics – India*. Kaggle. Available at: https://www.kaggle.com/datasets/nikhilmahajan29/crop-production-statistics-india (Accessed: 20 October 2025).
- Abhinand, A. (2020) *Crop Production in India*. Kaggle. Available at: https://www.kaggle.com/datasets/abhinand05/crop-production-in-india (Accessed: 20 October 2025).
- Breiman, L. (2001) 'Random forests', *Machine Learning*, 45(1), pp. 5–32. doi:10.1023/A:1010933404324
- Chen, T. and Guestrin, C. (2016) 'XGBoost: A scalable tree boosting system', *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 785–794. doi:10.1145/2939672.2939785.
- Dietterich, T.G. (2000) 'Ensemble methods in machine learning', *Multiple Classifier Systems*, 1857, pp. 1–15. doi:10.1007/3-540-45014-9_1.
- Jeong, J.H., Resop, J.P., Mueller, N.D., Fleisher, D.H., Yun, K., Butler, E.E., Timlin, D.J., Shim, K.M., Gerber, J.S., Reddy, V.R. and Kim, S.H. (2016) 'Random forests for global and regional crop yield predictions', *PLoS ONE*, 11(6), e0156571. doi:10.1371/journal.pone.0156571.
- Khaki, S. and Wang, L. (2019) 'Crop yield prediction using deep neural networks', *Frontiers in Plant Science*, 10, 621. doi:10.3389/fpls.2019.00621.
- Lobell, D.B. (2013) 'The use of satellite data for crop yield gap analysis', *Field Crops Research*, 143, pp. 56–64. doi:10.1016/j.fcr.2012.08.008.
- Lobell, D.B. and Burke, M.B. (2010) *Climate change and food security: Adapting agriculture to a warmer world*. Dordrecht: Springer. doi:10.1007/978-90-481-2953-9.
- Molnar, C. (2020) *Interpretable Machine Learning*. 2nd edn. Available at: https://christophm.github.io/interpretable-ml-book/ (Accessed: 25 October 2025).
- Muruganantham, P., Wibowo, S., Grandhi, S., Samrat, N.H. and Islam, N. (2022) 'A systematic literature review on crop yield prediction using remote sensing and machine learning', *Remote Sensing*, 14(9), 1990. doi:10.3390/rs14091990.
- Scikit-learn Developers (2023) *Scikit-learn: Machine learning in Python*. Available at: https://scikit-learn.org/stable/(Accessed: 22 October 2025).
- Shahhosseini, M., Hu, G. and Archontoulis, S.V. (2020) 'Forecasting corn yield with machine learning ensembles', *Frontiers in Plant Science*, 11, 1120. doi:10.3389/fpls.2020.01120.
- Van Klompenburg, T., Kassahun, A. and Catal, C. (2020) 'Crop yield prediction using machine learning: A systematic literature review', *Computers and Electronics in Agriculture*, 177, 105709.doi:10.1016/j.compag.2020.105709.

Appendices

Appendix A: Dataset Description and Summary Statistics

Purpose

This appendix provides supporting evidence for the dataset scale, structure, and variables used in the study, without interrupting the main methodological narrative.

Appendix A.1 Dataset Overview Table (TABLE 1)

Table A1: Overview of the Crop Production Dataset

Attribute	**Description**
Data source	Kaggle-Crop Production in India
Geographic scope	India(state and district level)
Temporal coverage	Multiple crop years
Total records (raw)	345,336
Records after cleaning	338,548
Number of features	8
Target variable	Yield (tonnes/hectare)
Data type	Structured tabular

Code:-

```
df_raw = pd.read_csv("APY.csv") print("Raw dataset shape:",
df_raw.shape)
df_raw.head()
df_raw.info()
print("Columns:", df_raw.columns.tolist())
print("\nMissing values per column:")
df_raw.isna().sum()
df_raw.describe()
 numeric = ["Crop_Year", "Area", "Production", "Yield"]
 fig, axes = plt.subplots(2, 2, figsize=(12, 8))
axes = axes.ravel() for i, col in
enumerate(numeric):
```

```
sns.histplot(df_before[col], bins=50, kde=True, ax=axes[i])
axes[i].set_title(f"{col} (Before Cleaning)") plt.tight_layout() plt.show()
plt.figure(figsize=(6,4))
sns.heatmap(df_before[numeric].corr(),annot=True, cmap="coolwarm")
plt.title("Correlation Before Outlier Removal")
plt.show()
outlier_cols = ["Area", "Production", "Yield"]
z = np.abs(zscore(df[outlier_cols]))
df_after = df[(z < 3).all(axis=1)].reset_index(drop=True)
print("Rows BEFORE outlier removal:", df_before.shape[0])
print("Rows AFTER outlier removal :", df_after.shape[0])
target = "Yield"
X = df_after.drop(columns=[target])
y = df_after[target]
X_train, X_test, y_train, y_test = train_test_split(
X, y, test_size=0.25, random_state=42
)
scale_cols = ["Area", "Production","Crop_Year"]
scaler = StandardScaler()
X_train_scaled = X_train.copy()
X_test_scaled = X_test.copy()
X_train_scaled[scale_cols] = scaler.fit_transform(X_train[scale_cols])
X_test_scaled[scale_cols] = scaler.transform(X_test[scale_cols])
print("Training shape:", X_train_scaled.shape)
print("Test shape :", X_test_scaled.shape)
plt.figure(figsize=(6, 4))
sns.heatmap(df_after[numeric].corr(),annot=True, cmap="coolwarm")
plt.title("Correlation After Outlier Removal")
plt.show()
season_median = (
df_after_plot
.groupby("Season_Name")["Yield"]
.median()
.reset_index()
)
plt.figure(figsize=(8,4))
sns.barplot(data=season_median, x="Season_Name", y="Yield")
plt.title("Median Yield Across Seasons")
plt.xlabel("Season")
plt.ylabel("Median Yield")
plt.xticks(rotation=30, ha="right")
plt.tight_layout()
plt.show()
season_mean = (
df_after_plot
.groupby("Season_Name")["Yield"]
.mean()
.reset_index()
)
plt.figure(figsize=(8,4))
sns.barplot(data=season_mean,x="Season_Name", y="Yield")
plt.title("Average Yield by Season")
```

```
plt.xlabel("Season")
plt.ylabel("Average Yield")
plt.xticks(rotation=30, ha="right")
plt.tight_layout()
plt.show()
state_stats = ( df_before
.groupby("State")["Yield"]
.mean()
.reset_index()
.sort_values("Yield", ascending=False)
.head(15)
)
plt.figure(figsize=(10,6))
sns.barplot(data=state_stats,x="Yield", y="State")
plt.title("Average Yield by State (Top 15)")
plt.xlabel("Average Yield")
plt.ylabel("State")
plt.tight_layout()
plt.show()
plt.figure(figsize=(10,5))
sns.countplot(
data=df_before,
x="Season",
order=df_before["Season"].value_counts().index
)
plt.title("Number of Records per Season")
plt.xlabel("Season")
plt.ylabel("Count")
plt.tight_layout()
plt.show()
from sklearn.metrics import r2_score, mean_absolute_error, mean_squared_error
import numpy as np
def evaluate_model(name, model, X_train, X_test, y_train, y_test):
model.fit(X_train, y_train)
preds = model.predict(X_test)
r2 = r2_score(y_test, preds)
mae = mean_absolute_error(y_test, preds)
rmse = np.sqrt(mean_squared_error(y_test, preds))
print(f"{name}: R2={r2:.4f}, MAE={mae:.4f},
RMSE={rmse:.4f}")
return [name, r2, mae, rmse]
from sklearn.linear_model import LinearRegression
lr = LinearRegression()
lr_result = evaluate_model("Linear Regression", lr,
X_train_scaled, X_test_scaled,
y_train, y_test)
lr_result_df = pd.DataFrame([lr_result], columns=["Model", "R2", "MAE",
"RMSE"])
lr_result_df
# 1. CREATE FULL RESULTS TABLE
results = [lr_result] + rf_results + et_results + hgb_results
```

```
results_df = pd.DataFrame(results, columns=["Model", "R2", "MAE",
"RMSE"])
# Extract model family name (part before underscore)
results_df['Family'] = results_df['Model'].apply(lambda x: x.split("_")[0])
# 2. FOR EACH FAMILY, KEEP THE BEST MODEL (highest R²)
best_per_family = (
results_df.loc[results_df.groupby("Family")["R2"].idxmax()]
.sort_values(by="R2", ascending=True)
.reset_index(drop=True)
)
# CLEAN final table (remove Family column)
final_df = best_per_family[["Model", "R2", "MAE", "RMSE"]]
print("=== BEST MODEL FROM EACH FAMILY ===")
display(final_df)
from sklearn.model_selection import cross_val_score
from sklearn.ensemble import
RandomForestRegressor
# Define the best RandomForest configuration (same as in your main
run)
best_rf = RandomForestRegressor(
 n_estimators=100,
max_depth=None,
random_state=42,
n_jobs=-1
)
# 5-fold cross-validation on the training data
cv_scores = cross_val_score(
best_rf,
X_train_scaled,
y_train, cv=5,
scoring="r2"
)
print("RandomForest 5-fold CV R2:")
print(" mean =", cv_scores.mean())
print(" std =", cv_scores.std())
Fit best RandomForest on the full training
set
best_rf = RandomForestRegressor(
n_estimators=100,
max_depth=None,
random_state=42,
n_jobs=-1
)
best_rf.fit(X_train_scaled, y_train)
# Compute and sort feature importances importances = best_rf.feature_importances_ feat_imp =
pd.Series(importances, index=X_train_scaled.columns).sort_values(ascending=False)
# Plot plt.figure(figsize=(8,4))
sns.barplot(x=feat_imp.values,
y=feat_imp.index) plt.title("Feature Importance -
Random Forest") plt.xlabel("Importance")
plt.ylabel("Feature") plt.tight_layout() plt.show()
```

Snippits

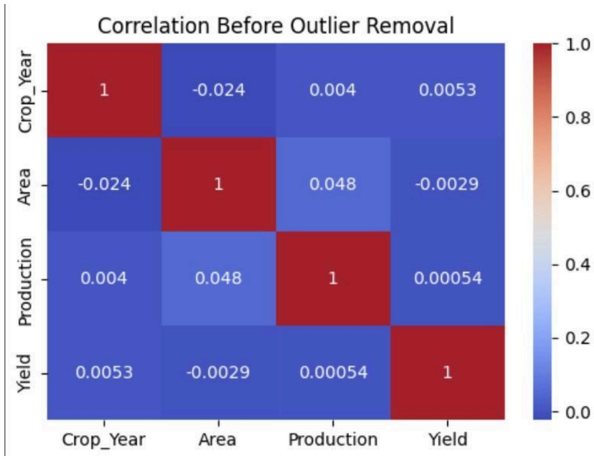

Figure 1. Correlation Structure Before Outlier Removal

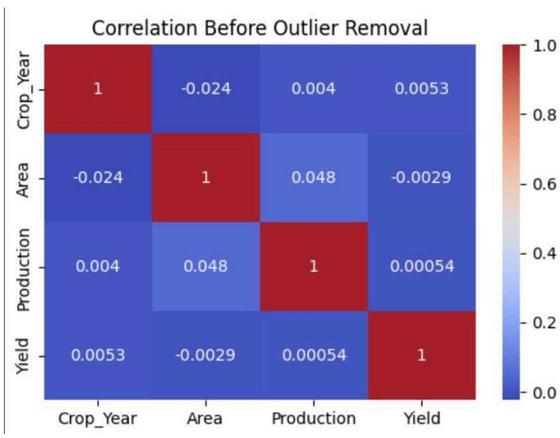

Figure 2. Correlation Structure After Outlier Removal

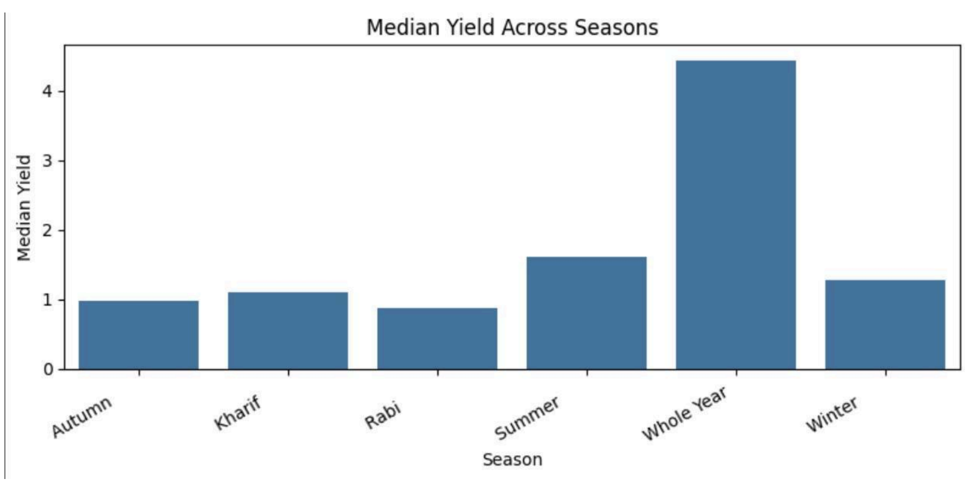

Figure 3. Median Crop Yield Variation Across Agricultural Seasons

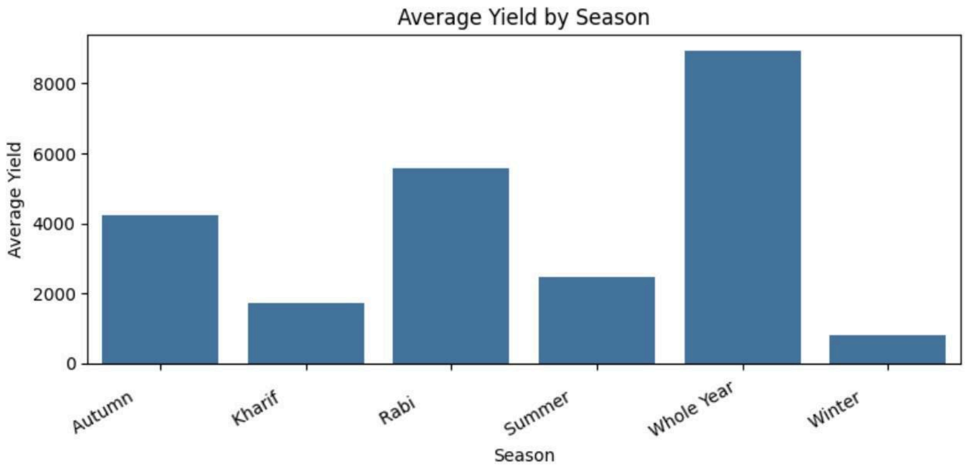

Figure 4. Average Crop Yield Across Agricultural Seasons

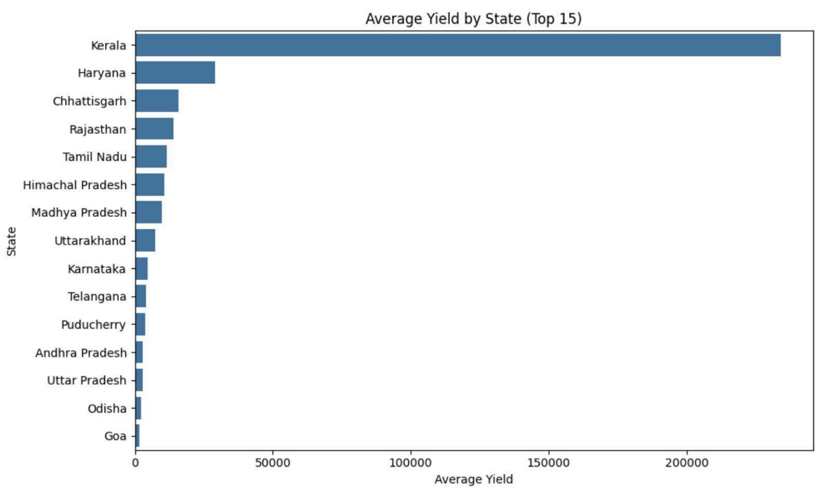

Figure 5. Average Crop Yield by State (Top 15 States)

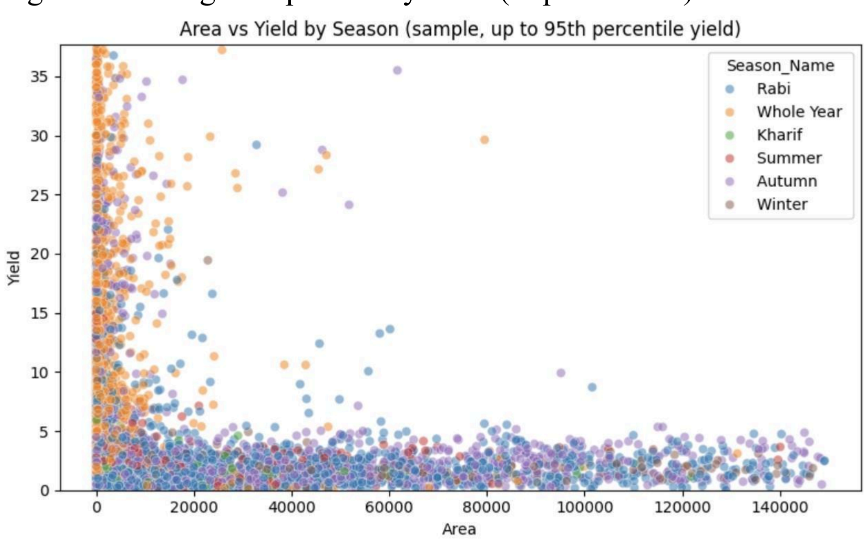

Figure 6. Non-Linear Relationship Between Cultivated Area and Yield Across Seasons

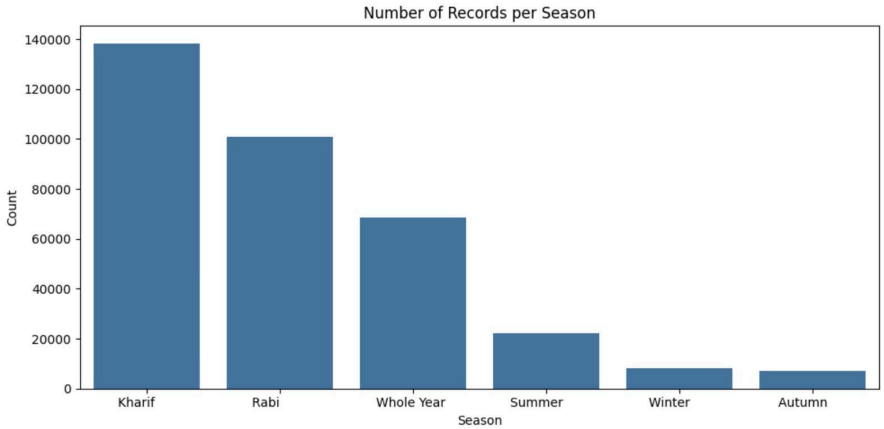

Figure 7. Distribution of Records Across Cropping Seasons

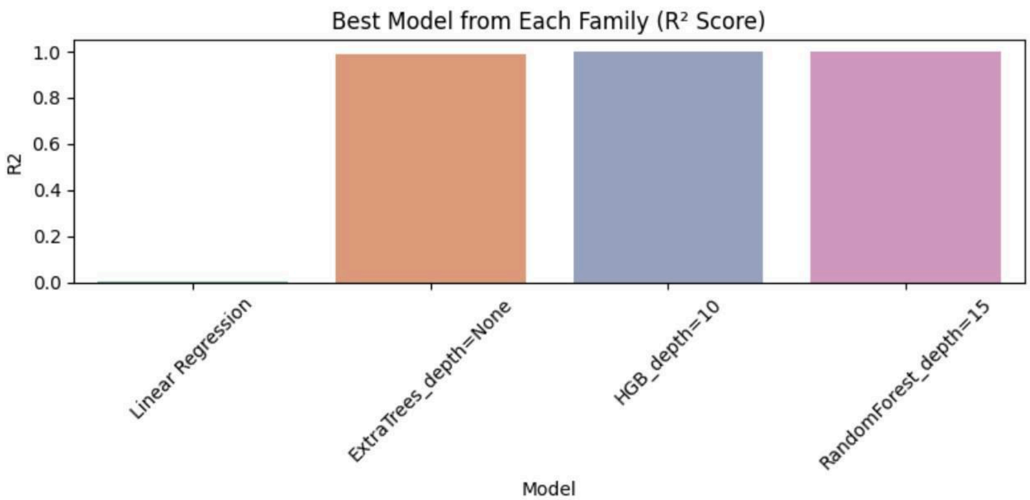

Figure 8. Best-Performing Model from Each Algorithm Family Based on R^2 Score

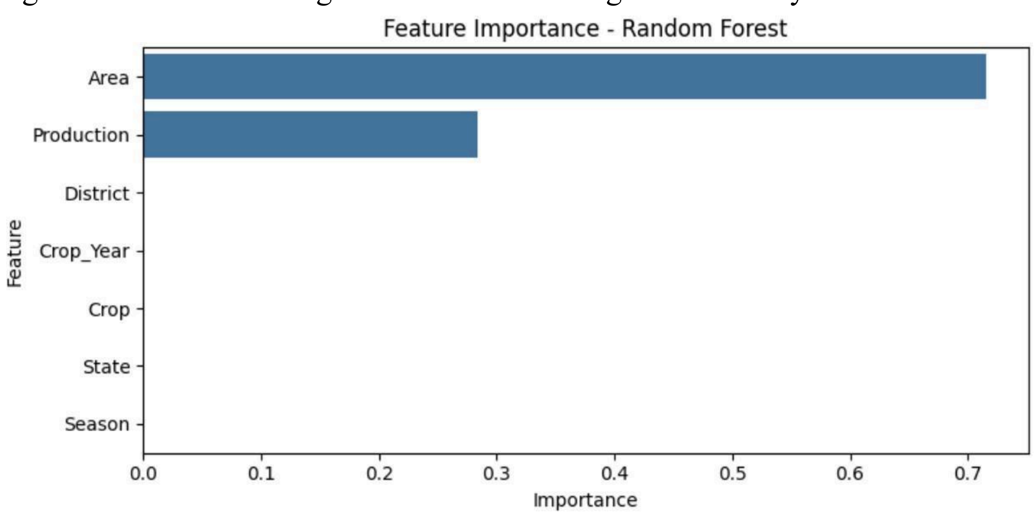

Figure 9. Feature Importance Ranking from the Best-Performing Random Forest Model